ReviewIQ

Aspect-Based Customer Review Analytics with Structured Outputs

5th Semester · AIML

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1 Problem Statement

E-commerce platforms receive large volumes of customer reviews that contain valuable product feedback. Because this feedback is unstructured, it is difficult to systematically identify customer sentiment, product-specific strengths and weaknesses, and recurring complaints. Traditional sentiment analysis assigns a single label to an entire review, overlooking the fact that a customer may hold different opinions about several aspects of the same product. The problem is to develop an LLM-based pipeline that transforms unstructured reviews into validated structured records containing overall sentiment, aspect-level sentiment and complaint categories, and that generates meaningful product-level summaries.

2 Why This Problem Matters

Reviews influence purchasing decisions and guide product improvement, yet analysing thousands of them by hand is slow, inconsistent and difficult to scale. A customer may praise a smartphone's camera and display but criticise its battery; a review-level classifier would simply label this as "mixed", hiding the actionable detail. Businesses need a system that identifies the aspects customers discuss, determines the sentiment towards each one, categorises recurring complaints, summarises feedback per product, and produces reliable, machine-readable output for further analysis.

3 Proposed Solution

ReviewIQ is an LLM-powered review analytics application that combines prompt engineering, schema-based structured outputs and automatic validation to convert raw e-commerce reviews into actionable product intelligence. The language model is treated as an unreliable component: every output is validated against a Pydantic schema, its evidence is checked against the original review text, and invalid output is repaired or rejected rather than silently accepted. Results are explored through an interactive Streamlit dashboard and can be exported as JSON and CSV.

4 Key Components of the Solution

4.1 Structured Review Extraction

LangChain prompts and Pydantic schemas extract the overall sentiment (positive, negative, neutral or mixed), product aspects with aspect-level sentiment and a verbatim supporting quote, complaint categories and a confidence score, using a fixed taxonomy of 13 aspect and 14 complaint categories.

4.2 Output Validation and Repair

Generated JSON is parsed and validated against the schema, and all evidence must be grounded in the review text. Malformed output first receives deterministic local fixes, then up to two configurable LLM repair attempts; if it remains invalid, a controlled failure is recorded. Every attempt is logged.

4.3 Prompt Engineering Comparison

Zero-shot, few-shot and Chain-of-Thought prompts share one model, schema and set of label definitions for a fair comparison. The system is designed to evaluate them against 300 manually labelled reviews, with macro F1-score as the primary metric.

4.4 Product-Level Aggregation

Validated records are grouped by product with pandas to compute sentiment distributions, the most frequently discussed aspects and their positive–negative balance, recurring complaint categories, and representative evidence quotes.

4.5 Automated Product Summarization

An extract → aggregate → summarise prompt chain produces concise product summaries from the computed statistics. Each point of praise or complaint must cite valid supporting review IDs, and the model may not invent numbers.

4.6 LLM-as-a-Judge Evaluation

A judge model scores each summary on an anchored 1–5 rubric for faithfulness, coverage, consistency and unsupported claims, complemented by deterministic checks for invented figures and invalid citations.

5 Expected Outcome

A functional review analytics platform that converts unstructured e-commerce feedback into validated, structured information and interpretable product-level insights, accompanied by a comparative evaluation of zero-shot, few-shot and Chain-of-Thought prompting. The project will demonstrate how LLMs can be applied to structured information extraction, multi-aspect sentiment analysis and automated summarisation.

6 Technology Stack

Language and data	Python, Pandas, JSON and CSV; public Amazon Reviews 2023 data (Appliances category)
LLM orchestration	LangChain prompt templates and output parsers; Pydantic for schema validation
LLM provider	Google Gemini as the planned, configurable provider, behind a provider-independent interface
Evaluation	Scikit-learn for precision, recall, macro F1 and confusion matrices
Interface	Streamlit dashboard with Plotly visualisations

Project status. The pipeline and dashboard are implemented and tested offline. Live experiments with Gemini and the manual annotation of the 300-review evaluation set are still pending; accordingly, no experimental results are reported in this proposal.